

1

RM-Class-13-14



Class-13-14: Testing of Hypothesis

2

Learning Objectives of Class-13

By the end of this class the student must be able to answer the following questions.

- Define Null Hypothesis and Alternate Hypothesis
- What are the Type-1 and Type-2 Error
- How would you test of the significance of the mean of the single population using both t and Z-test?
- How would you test of the significance of difference between two population means using both t and Z-test?
- How you will test the significance of single population proportion?
- How you will test the equality of two population proportions?

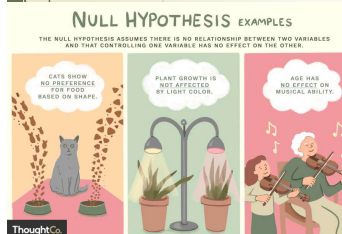
3

Define Null Hypothesis and Alternate Hypothesis

4

Null Hypothesis

- Null hypothesis (H_0) is a statement in statistics that assumes there is no effect, no difference, or no relationship between variables
- In hypothesis testing, the null hypothesis assumes no relationship between two variables, providing a baseline for statistical analysis.
- Rejecting the null hypothesis suggests there is evidence of a relationship between variables.
- By formulating a null hypothesis, researchers can systematically test assumptions and draw more reliable conclusions from their experiments.
- A small p-value (usually $p < 0.05$) provides evidence against the null hypothesis, leading researchers to reject H_0 . A large p-value means there is not enough evidence to reject it.



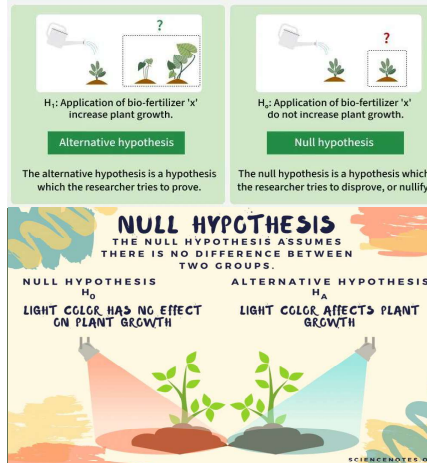
5

Example: Null Hypothesis

Question	Null Hypothesis
Are teens better at math than adults?	Age has no effect on mathematical ability.
Does taking aspirin every day reduce the chance of having a heart attack?	Taking aspirin daily does not affect heart attack risk.
Do teens use cell phones to access the internet more than adults?	Age has no effect on how cell phones are used for internet access.
Do cats care about the color of their food?	Cats express no food preference based on color.
Does chewing willow bark relieve pain?	There is no difference in pain relief after chewing willow bark versus taking a placebo.

6

Alternate Hypothesis



- The **alternative hypothesis (H_a)** is a statement in statistical hypothesis testing that suggests there is a **significant difference, effect, or relationship** in the population in terms of independent and dependent variables.
- The alternative hypothesis is the opposite of the null hypothesis and is accepted when there is enough statistical evidence against H_0 . Rejection of null hypothesis leads to the acceptance of alternate hypothesis.

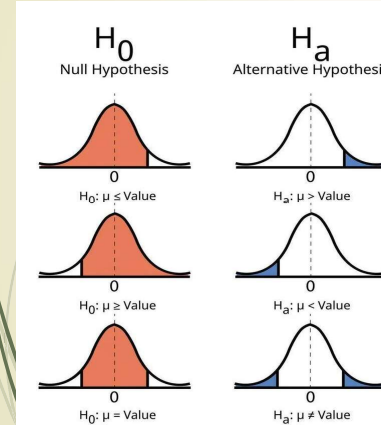
7

Example

Question / Problem	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)
Are teens better at math than adults?	Age has no effect on mathematical ability.	Teens have better mathematical ability than adults.
Does taking aspirin every day reduce the chance of having a heart attack?	Taking aspirin daily does not affect heart attack risk.	Taking aspirin daily reduces the risk of having a heart attack.
Do teens use cell phones to access the internet more than adults?	Age has no effect on how cell phones are used for internet access.	Teens use cell phones to access the internet more than adults.
Do cats care about the color of their food?	Cats show no food preference based on color.	Cats show a preference for certain food colors.
Does chewing willow bark relieve pain?	There is no difference in pain relief after chewing willow bark compared with taking a placebo.	Chewing willow bark provides greater pain relief compared with taking a placebo.

8

One-Sided and Two-Sided Hypothesis Tests



- A **one-sided test** is used when the alternative hypothesis predicts a direction of change.
- It has **one rejection region** in one tail of the distribution.

Types:

(i) Right-tailed test

Used when: $H_a: \mu > \text{Value}$

Example:

- $H_0: \mu \leq 100$
- $H_a: \mu > 100$

The critical region is on the **right side**.

(ii) Left-tailed test

Used when: $H_a: \mu < \text{Value}$

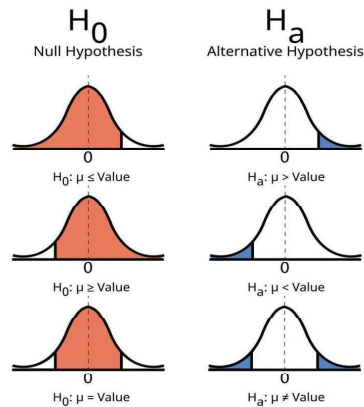
Example:

- $H_0: \mu \geq 100$
- $H_a: \mu < 100$

The critical region is on the **left side**.

9

One-Sided and Two-Sided Hypothesis Tests

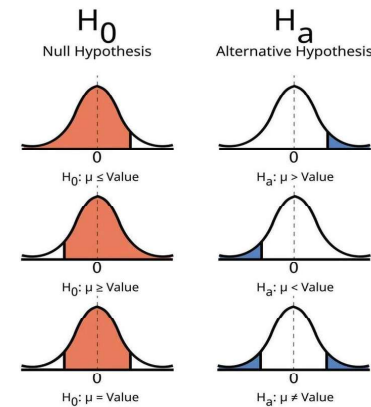


B. Two-Sided Hypothesis Test (Two-Tailed Test)

- A two-sided test is used when we are interested in whether the value is simply different, without specifying direction.
- $H_a: \mu \neq \text{"Value"}$
- Example:
 - $H_0: \mu = 100$
 - $H_a: \mu \neq 100$
- The rejection regions are placed in both tails of the distribution.

10

Critical Analysis on one-sided and two-sided hypothesis tests



Meaning of the Shaded Areas

- Blue shaded area = rejection (critical) region.
- Unshaded area = acceptance/failure-to-reject region.
- The boundary between them is called the critical value.
- If the calculated test statistic enters the shaded region → reject H_0 .
- If it stays outside → fail to reject H_0 .

In summary:

- $H_a > \text{Value} \rightarrow \text{Right-tailed test}$
- $H_a < \text{Value} \rightarrow \text{Left-tailed test}$
- $H_a \neq \text{Value} \rightarrow \text{Two-tailed test}$

11

What are the Type-1 and Type-2 Error

12

Type-I error

- Definition:** A Type-I error occurs when the researcher rejects the null hypothesis (H_0) even though it is actually true.
- In simple words: **It is rejecting a true null hypothesis.**
- Example:** Research question: Does a new medicine cure a disease?
Actual situation: The medicine does not work (H_0 is true).
Research conclusion: The medicine works (reject H_0).
The researcher has made a Type-I error.
- Probability:** Type-I error is represented by a (α). It is termed as level of significance.
Usually, $\alpha = 0.05$ (5% chance of making this error).

13

Type-II error

➤ **Definition:** A Type-II error occurs when the researcher fails to reject the null hypothesis (H_0) even though it is false.

In simple words: **It is accepting or not rejecting a false null hypothesis.**

➤ **Example:** Research question: Does a new medicine cure a disease?

Actual situation: The medicine works (H_0 is false).

Research conclusion: The medicine does not work (fail to reject H_0).

The researcher has made a Type-II error.

➤ **Probability:** Type-II error is represented by β (beta). The term $(1 - \beta)$ is termed as power of tests.

14

Decision Rule

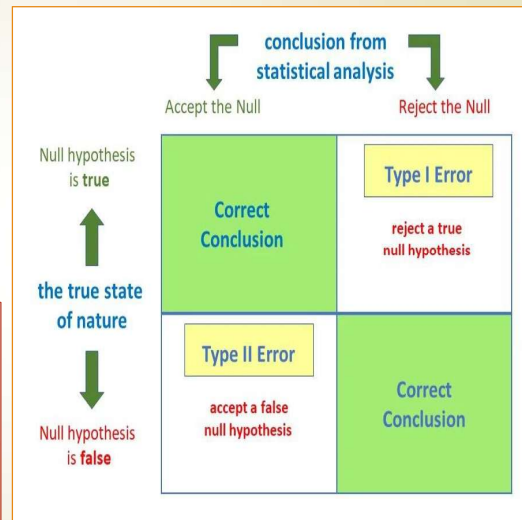
Actual Situation	Research Decision	Result
H_0 is true	Reject H_0	Type-I Error
H_0 is true	Fail to reject H_0	Correct decision
H_0 is false	Reject H_0	Correct decision
H_0 is false	Fail to reject H_0	Type-II Error

Type-I Error: "I reject H_0 , but H_0 was actually true."
Type-II Error: "I accept H_0 , but H_0 was actually false."

15

	True state of the world	
Decision	H_0 False	H_0 True
Reject H_0	Correct decision	Type I error
Keep H_0	Type II error	Correct decision

	Null Hypothesis is TRUE	Null Hypothesis is FALSE
Reject null hypothesis	Type I Error (False positive)	Correct Outcome! (True positive)
Fail to reject null hypothesis	Correct Outcome! (True negative)	Type II Error (False negative)



16

How would you test of the significance of the mean of the single population using both t and Z-test?

Steps in testing of hypothesis exercise

17

The main **steps in testing a hypothesis** are:

- 1. State the hypotheses**
 - Null hypothesis: H_0
 - Alternative hypothesis: H_1
- 2. Choose the level of significance:** Usually $\alpha = 0.05$ or 0.01 .
- 3. Select the appropriate statistical test:** For example: Z-test, t-test, chi-square test, or F-test, depending on the data and problem.
- 4. Calculate the test statistic:** Use the sample data and the relevant formula.
- 5. Find the critical value or p-value:** Use the significance level and the appropriate statistical table or software.
- 6. Make the decision**
 - If **p-value** $\leq \alpha$, reject H_0 .
 - If **p-value** $> \alpha$, fail to reject H_0 .
- 7. Write the conclusion:** State the result in simple words related to the original problem.

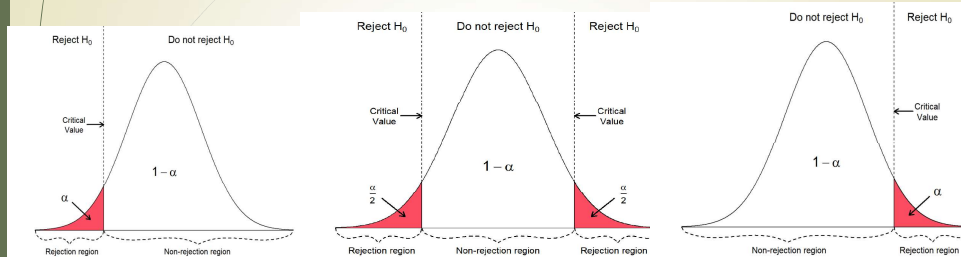
Level of significance

18

- The **level of significance**, represented by α , is the **maximum probability of rejecting the null hypothesis when it is actually true.**
- Common significance levels are: **5% = 0.05, 1% = 0.01**
- For example, if $\alpha = 0.05$, we accept a 5% risk of making a wrong decision by rejecting a true null hypothesis.
- In most statistical problems, the significance level is given in the question. If it is not given, 5% is commonly used.**
- A 5% level of significance means that there are 5 chances out of hundred that a null hypothesis will get rejected when it should be accepted.

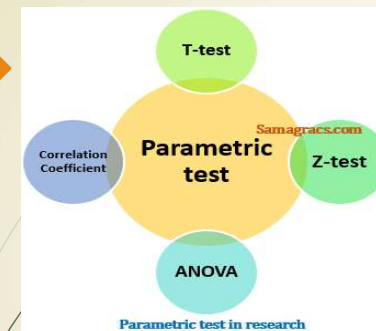
Determination of critical region with critical value (P value) and significance level (alpha)

19

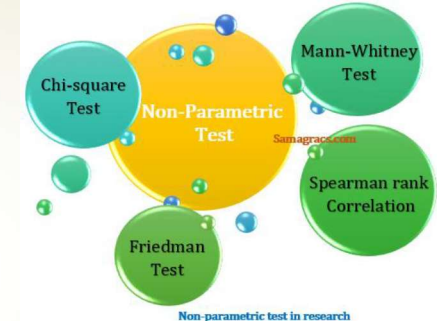


Note: if $p \leq \alpha$, reject H_0 ; otherwise, if $p > \alpha$, do not reject H_0

20



A parametric test is a statistical test based on assumptions about population parameters such as the mean and standard deviation.



A non-parametric test is a statistical test that does not depend on specific assumptions about the population distribution.

21

Testing the Significance of the Mean of a Single Population Using Z-Test and t-Test

In research methodology, a **test of significance of the population mean** is used to determine whether the observed sample mean is significantly different from a hypothesized population mean.

A researcher usually starts with a **null hypothesis (H_0)** that states there is **no significant difference**, and an **alternative hypothesis (H_a)** that states a significant difference exists.

Example research question: "Is the average weight of students different from the national average of 60 kg?"

Here:

- Population mean (hypothesized) = 60 kg
- Sample mean is calculated from collected data
- Statistical testing determines whether the difference is due to chance or a real effect.

22

Hypotheses for Testing a Single Population Mean

Suppose the population mean is represented by μ_0 .

Null Hypothesis (H_0): $H_0: \mu = \mu_0$

Meaning: The population mean is equal to the assumed value.

Alternative Hypothesis (H_a):

Three possible forms:

(A) Two-tailed test: $H_a: \mu \neq \mu_0$

Used when we want to know whether the mean is simply different.

(B) Right-tailed test: $H_a: \mu > \mu_0$

Used when we expect an increase.

(C) Left-tailed test: $H_a: \mu < \mu_0$

Used when we expect a decrease.

23

Z-Test for Single Population Mean

► **Definition of Z-Test:** A **Z-test for a single population mean** is a statistical test used to determine whether a sample mean differs significantly from a known population mean when the **population standard deviation is known** or the sample size is large.

► **When to Use Z-Test?**

Use a Z-test when:

Condition	Explanation
Population standard deviation (σ) is known	The true population variability is available
Sample size is large	Usually $n \geq 30$
Population distribution is approximately normal	Data follows normal distribution
Observations are independent	One observation does not influence another

24

Z-Test Procedure (Steps)

Step 1: State the Hypotheses

► Null hypothesis:

$$H_0: \mu = \mu_0$$

(The sample comes from a population with mean μ_0)

► Alternative hypothesis:

$$H_1: \mu \neq \mu_0$$

(or $H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$)

Step 2: Select the Level of Significance

Usually:

$$\alpha = 0.05$$

or

$$\alpha = 0.01$$

25

Continue....

Step 3: Calculate the Test Statistic

Use:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Symbol	Meaning
Z	Calculated Z-test statistic
$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
σ	Population standard deviation
n	Sample size
$\sigma/\sqrt{n}$	Standard error of the mean

26

Continue...

Step 4: Find the Critical Value

For example, at 5% significance:

Two-tailed test:

$$Z_{0.05} = \pm 1.96$$

One-tailed test:

$$Z_{0.05} = \pm 1.645$$

Step 5: Decision Rule

- If calculated Z lies in the rejection region → Reject H_0 .
- If calculated Z lies in the acceptance region → Fail to reject H_0 .

27

Example of Z-Test

Problem: A company claims that the average lifetime of a bulb is **1000 hours**.

A sample of **50 bulbs** has:

- ☑ Sample mean = 1020 hours
- ☐ Population standard deviation = 80 hours
- ☐ Significance level = 5%

Test whether the bulb life is different from 1000 hours.

28

Continue...

Step 1: Hypothesis: $H_0: \mu = 1000$;
 $H_a: \mu \neq 1000$

(two-tailed test)

Step 2: Formula:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Substitute values:

$$Z = \frac{1020 - 1000}{80/\sqrt{50}}$$

$$Z = \frac{20}{80/7.07}$$

$$Z = \frac{20}{11.31}$$

$$Z = 1.77$$

Step 3: Decision: Critical value:
 ± 1.96 (Calculate from the table-
Annexures-1)

Calculated: $Z = 1.77$

Since: $1.77 < 1.96$

We fail to reject H_0 .

Conclusion: There is not enough evidence to say that the bulb life is different from 1000 hours.

t-Test for Single Population Mean

29

- **Definition of t-Test:** A **one-sample t-test** is used to test whether a sample mean significantly differs from a population mean when the **population standard deviation is unknown** and estimated using the sample standard deviation.

➤ When to Use t-Test?

Use a t-test when:

Condition	Explanation
Population standard deviation is unknown	σ is unavailable
Sample standard deviation is used	Use s instead of σ
Small sample size	Usually $n < 30$
Population is approximately normally distributed	Normality assumption required

t-Test Procedure

30

Step 1: State Hypotheses

Example: $H_0: \mu = 50$; $H_a: \mu > 50$

Step 2: Choose Significance Level

Usually: $\alpha = 0.05$

Step 3: Calculate t-value

Formula:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Symbol	Meaning
t	Calculated t-statistic
$\bar{X}$	Sample mean
μ_0	Hypothesized population mean
s	Sample standard deviation
n	Sample size
$s/\sqrt{n}$	Standard error
df	Degrees of freedom

Degrees of freedom:

$$df = n - 1$$

31

Continue....

Step 4: Find Critical t-value

Use the t-table with:

$$df = n - 1$$

Step 5: Compare and Decide

- If calculated t is greater than the critical value $\rightarrow$ Reject H_0 .
- Otherwise $\rightarrow$ Fail to reject H_0 .

32

Example of t-Test

Problem: A researcher claims that the average exam score of students is **70 marks**.

A sample of 10 students gives:

- ☐ Sample mean = 75
- ☐ Sample standard deviation = 8
- ☐ $\alpha = 0.05$

Test the claim

33

Step 1: Hypothesis: $H_0: \mu = 70$;
 $H_a: \mu \neq 70$

Two-sided test

Step 2: Calculate t-value

Formula: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Substitute:

$$t = \frac{75 - 70}{8/\sqrt{10}}$$

$$t = \frac{5}{8/3.16}$$

$$t = \frac{2.53}{5}$$

$$t = 1.98$$

Step 3: Degrees of Freedom

$$df = n - 1$$

$$df = 10 - 1 = 9$$

At $\alpha = 0.05$ (two-tailed):

$t_{critical} = 2.262$ (Calculated from Annexure-2)

Step 4: Decision:

Calculated:

$$t = 1.98$$

Critical:

$$2.262$$

Since:

$$1.98 < 2.262$$

Fail to reject H_0 .

Conclusion: There is insufficient evidence to conclude that the average exam score differs from 70 marks.

T Test vs Z Test: What Is the Difference?

34

- ❑ Known population standard deviation → Use Z-test
- ❑ Unknown population standard deviation → Use one-sample t-test

	T TEST	Z TEST
		
	t-distribution	Standard normal (Z) distribution
Σ Population standard deviation	Unknown	Known
Sample size	Small (typically $n < 30$)	Large (typically $n \geq 30$)
Distribution used	t-distribution	Standard normal (Z) distribution
df Degrees of freedom	df = n - 1	Not applicable
Margin of error	Wider (t critical value is larger)	Narrower (Z critical value is smaller)
Common use	Small samples, unknown population σ	Large samples, known population σ

35

How would you test of the significance of difference between two population means using both t ad Z-test?

36

Testing the Significance of the Difference Between Two Population Means Using Z-Test and t-Test

In research methodology, **a test of significance between two population means** is used to determine whether the difference observed between two sample means is statistically significant or whether it occurred due to random chance.

It is commonly used when comparing two groups, such as:

- Effectiveness of two teaching methods
- Difference in blood pressure between two treatments
- Difference in income between two populations
- Comparison of performance between two groups

The objective is to test whether: $\mu_1 - \mu_2 = 0$

where there is **no real difference** between the two population means.

Hypotheses for Comparing Two Population Means

37

Let:

- μ_1 = Mean of population 1
- μ_2 = Mean of population 2

Null Hypothesis (H_0): The null hypothesis states that there is **no significant difference** between the two population means.

$$H_0: \mu_1 = \mu_2$$

or

$$H_0: \mu_1 - \mu_2 = 0$$

Meaning: The difference between the two populations is due to chance.

Alternative Hypothesis (H_a): The alternative hypothesis states that a significant difference exists.

Two-tailed test:

$$H_a: \mu_1 \neq \mu_2$$

Right-tailed test:

$$H_a: \mu_1 > \mu_2$$

Left-tailed test:

$$H_a: \mu_1 < \mu_2$$

Part a: z-test for difference between two population means

38

- **Definition of Two-Sample Z-Test:** A **two-sample Z-test** is a statistical test used to determine whether the difference between two sample means is significant when the **population standard deviations are known** and the samples are usually large.

- It compares: $\bar{X}_1 - \bar{X}_2$

with the expected difference: $\mu_1 - \mu_2$

- **When to Use Z-Test?:** A two-sample Z-test is used when:

Condition	Explanation
Population standard deviations are known	σ_1 and σ_2 are available
Sample sizes are large	Usually $n_1 \geq 30$ and $n_2 \geq 30$
Data follows approximately normal distribution	Normality assumption
Samples are independent	One sample does not influence another

Formula for Two Population Mean Z-Test

39

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Under H_0 :

$$\mu_1 - \mu_2 = 0$$

Therefore:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Symbol	Meaning
Z	Calculated Z-test statistic
$\bar{X}_1$	Mean of sample 1
$\bar{X}_2$	Mean of sample 2
μ_1	Population mean of group 1
μ_2	Population mean of group 2
σ_1	Population standard deviation of group 1
σ_2	Population standard deviation of group 2
n_1	Sample size of group 1
n_2	Sample size of group 2

Procedure of Z-Test

40

Step 1: State Hypotheses

Example:

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

Step 2: Select Significance Level

Usually:

$$\alpha = 0.05$$

Step 3: Calculate Z Value

Using the formula.

Step 4: Find Critical Value

For $\alpha = 0.05$:

Two-tailed:

$$Z_c = \pm 1.96$$

For a one-tailed test: it is $Z_c = \pm 1.645$

Step 5: Decision Rule

- ☐ If calculated Z > critical value $\rightarrow$ Reject H_0
- ☐ If calculated Z falls within acceptance region $\rightarrow$ Fail to reject H_0

41

Example of Two-Sample Z-Test

Problem:

A researcher wants to compare the average blood pressure of two groups.

Group A:

- Sample size (n_1) = 50
- Mean ($\bar{X}_1$) = 130 mmHg
- Population SD (σ_1) = 15

Group B:

- Sample size (n_2) = 60
- Mean ($\bar{X}_2$) = 125 mmHg
- Population SD (σ_2) = 18

Test whether there is a significant difference.

42

Example of Two-Sample Z-Test

Step 1: Hypothesis

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

Step 2: Calculate Z

$$\begin{aligned} Z &= \frac{130 - 125}{\sqrt{\frac{15^2}{50} + \frac{18^2}{60}}} \\ &= \frac{5}{\sqrt{4.5 + 5.4}} \\ &= \frac{5}{3.15} \\ Z &= 1.59 \end{aligned}$$

Step 3: Decision

Critical value:

$$\pm 1.96$$

Calculated value:

$$1.59$$

Since:

$$1.59 < 1.96$$

Conclusion:

There is no significant difference between the two population means.

43

Part b: t-test for difference between two population means

Definition of Two-Sample t-Test

A **two-sample t-test** is used to determine whether two independent sample means are significantly different when the **population standard deviations are unknown**.

The sample standard deviations are used instead.

When to Use t-Test?

Condition	Explanation
Population SD unknown	σ_1 and σ_2 are unavailable
Sample sizes are small	Usually $n_1 < 30$ or $n_2 < 30$
Sample SD is available	Use s_1 and s_2
Data approximately normal	Normality assumption

44

Formula for Independent Two-Sample t-Test

$$t = \frac{X_1 - X_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Where pooled variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Degrees of freedom:
df = $n_1 + n_2 - 2$

Symbol	Meaning
t	Calculated t-test statistic
$\bar{X}_1$	Mean of sample 1
$\bar{X}_2$	Mean of sample 2
s_1	Sample SD of group 1
s_2	Sample SD of group 2
n_1	Sample size of group 1
n_2	Sample size of group 2
s_p^2	Pooled variance
df	Degrees of freedom

45

Procedure of Two-Sample t-Test

Step 1:

State H_0 and H_a .

Step 2:

Select significance level.

Example: $\alpha = 0.05$

Step 3:

Calculate pooled variance.

Step 4:

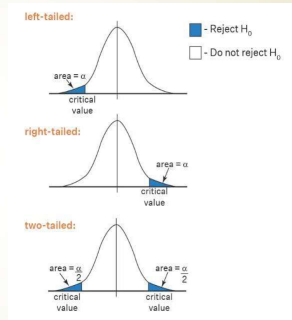
Calculate t-value.

Step 5:

Compare with critical t-value. **The critical t value is calculated from Annexure-II**

Step 6:

Accept or reject H_0 .



46

Example of Two-Sample t-Test

Problem:

Compare exam scores between two teaching methods.

Method A:

- $n_1 = 10$
- Mean = 80
- SD = 6

Method B:

- $n_2 = 12$
- Mean = 75
- SD = 5

47

Continue...

Hypothesis

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

Calculate pooled variance:

$$\begin{aligned} s_p^2 &= \frac{(10-1)(6^2) + (12-1)(5^2)}{10+12-2} \\ &= \frac{9(36) + 11(25)}{20} \\ &= \frac{324 + 275}{20} \\ &= 29.95 \end{aligned}$$

Calculate t-value:

$$\begin{aligned} t &= \frac{80 - 75}{\sqrt{29.95 \left(\frac{1}{10} + \frac{1}{12} \right)}} \\ &= \frac{5}{\sqrt{5.49}} \\ &= 2.13 \end{aligned}$$

Degrees of freedom: $df = 10 + 12 - 2 = 20$

Critical t at $\alpha=0.05$:

$t_c = 2.086$ (From Annexure-II)

Decision:

Calculated:

$$t = 2.13$$

Critical:

$$2.086$$

Since:

$$2.13 > 2.086$$

Reject H_0 .

Conclusion:

There is a significant difference between the two teaching methods.

48

Summary Decision Rule

Key concept:

- Z-test compares two means using known population variability.
- t-test compares two means when variability must be estimated from samples.

Situation	Appropriate Test
Comparing two means + population SD known + large samples	Two-sample Z-test
Comparing two means + population SD unknown + small samples	Independent two-sample t-test

How you will test the significance of single population proportion?

Testing the Significance of a Single Population Proportion (One-Sample Proportion Test)

- A single population proportion test is used to determine whether the proportion of a population is significantly different from a specified value.
- For example:
 - Testing whether the proportion of defective products is 5%.
 - Testing whether the proportion of voters supporting a candidate is 50%.
 - Testing whether a disease affects a certain percentage of people.

Stepwise Procedure

Step 1: State the Hypotheses

Let:

- p = true population proportion
- p_0 = hypothesized population proportion

Null hypothesis:

$$H_0: p = p_0$$

Alternative hypothesis:

Depending on the problem:

- Two-tailed test:

$$H_1: p \neq p_0$$

- Right-tailed test:

$$H_1: p > p_0$$

- Left-tailed test:

$$H_1: p < p_0$$

Continue....

Step 2: Select the Level of Significance

- Choose the significance level (α). Common choices: $\alpha=0.05$ "or" $\alpha=0.01$

Step 3: Calculate Sample Proportion

- The sample proportion is: $\hat{p} = \frac{x}{n}$
- where: x = number of successes, n = sample size

Step 4: Calculate the Test Statistic

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- $\hat{p}$ = sample proportion
- p_0 = hypothesized population proportion
- n = sample size

Continue....

53

Step 5: Find the Critical Value or p-value

Use the standard normal distribution table.

Common critical values:

Significance Level	Two-tailed Critical Value
5% ($\alpha = 0.05$)	± 1.96
1% ($\alpha = 0.01$)	± 2.58

For one-tailed tests:

□ Right-tailed at 5%: $Z = 1.645$

□ Left-tailed at 5%: $Z = -1.645$

Step 6: Decision Rule

□ If calculated Zvalue falls in the rejection region → Reject H_0 .

□ If calculated Zvalue does not fall in the rejection region → Fail to reject H_0 .

Step 7: State the Conclusion

Write the conclusion in words related to the problem.

Example

54

Problem: A company claims that **80% of its customers are satisfied** with its service. A survey of **100 customers** shows that **72 customers are satisfied**. Test whether the company's claim is correct at a **5% significance level**.

Step 1: State Hypotheses

Claim:

$$p_0 = 0.80$$

Null hypothesis:

$$H_0: p = 0.80$$

Alternative hypothesis:

$$H_1: p \neq 0.80$$

(We use a two-tailed test because we are checking whether the claim is different.)

Continue...

55

Step 3: Calculate Sample Proportion

Given: $x = 72, n = 100$

$$\hat{p} = \frac{72}{100}$$

$$\hat{p} = 0.72$$

Step 4: Calculate Test Statistic

Formula:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Substitute values:

$$Z = \frac{0.72 - 0.80}{\sqrt{\frac{0.80(1-0.80)}{100}}}$$

$$Z = \frac{-0.08}{\sqrt{\frac{0.16}{100}}}$$

$$Z = \frac{-0.08}{0.04}$$

$$Z = -2$$

Continue...

56

Step 5: Compare with Critical Value

Calculated: $Z = -2$

Critical values: -1.96 and $+1.96$

Since: $-2 < -1.96$

the calculated value lies in the rejection region.

Step 6: Decision

Reject H_0 .

Step 7: Conclusion

At the 5% significance level, there is sufficient statistical evidence to conclude that the true proportion of satisfied customers is **different from 80%**.

57

How you will test the significance of two population proportion?

58

Test of Significance of Two Population Proportions (Two-Proportion Z-Test)

A **two-population proportion test** is used to determine whether there is a significant difference between the proportions of two independent populations.

It is used when comparing:

- Proportion of defective items from two factories.
- Success rates of two different treatments.
- Customer satisfaction rates of two companies.

The test compares two population proportions: p_1 and p_2

59

Stepwise Procedure

Step 1: State the Hypotheses

Let:

- p_1 = proportion of population 1
- p_2 = proportion of population 2

Null Hypothesis:

$$H_0: p_1 = p_2$$

or

$$H_0: p_1 - p_2 = 0$$

Alternative Hypothesis:

Two-tailed test:

$$H_1: p_1 \neq p_2$$

Right-tailed test:

$$H_1: p_1 > p_2$$

Left-tailed test:

$$H_1: p_1 < p_2$$

60

Continue...

Step 2: Select the Level of Significance

Choose the significance level:

$$\alpha = 0.05 \text{ or } \alpha = 0.01$$

Step 3: Calculate Sample Proportions

For population 1:

$$\hat{p}_1 = \frac{x_1}{n_1}$$

For population 2:

$$\hat{p}_2 = \frac{x_2}{n_2}$$

where:

- x_1, x_2 = number of successes
- n_1, n_2 = sample sizes

Step 4: Calculate the Pooled Proportion

Under $H_0: p_1 = p_2$, we assume both populations have the same proportion.

The pooled proportion is:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Continue....

61

Step 5: Calculate the Test Statistic

The Z-test statistic is:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Step 6: Find Critical Value or p-value

For a 5% significance level:

Two-tailed test:

$$Z = \pm 1.96$$

One-tailed test:

$$Z = 1.645 \text{ or } -1.645$$

Step 7: Make Decision

- If calculated Zvalue falls in the rejection region → **Reject** H_0 .
- If calculated Zvalue does not fall in the rejection region → **Fail to reject** H_0 .

Step 8: State the Conclusion

Interpret the result in the context of the problem.

Example

62

► Problem: Two companies advertise their products.

A survey gives the following results:

Company	Customers Surveyed	Customers Satisfied
A	200	150
B	250	165

Test whether there is a significant difference between the satisfaction proportions of the two companies at a **5% significance level**.

Solution

63

Step 1: State Hypotheses: Let:

- p_1 = satisfaction proportion of Company A
- p_2 = satisfaction proportion of Company B

Null hypothesis:

$$H_0: p_1 = p_2$$

Alternative hypothesis:

$$H_1: p_1 \neq p_2$$

(Two-tailed test)

Step 2: Select Significance Level

$$\alpha = 0.05$$

Critical value:

$$Z = \pm 1.96$$

Step 3: Calculate Sample Proportions

For Company A:

$$\hat{p}_1 = \frac{150}{200}$$

$$\hat{p}_1 = 0.75$$

For Company B:

$$\hat{p}_2 = \frac{165}{250}$$

$$\hat{p}_2 = 0.66$$

64

Step 4: Calculate Pooled Proportion

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$\hat{p} = \frac{150 + 165}{200 + 250}$$

$$\hat{p} = \frac{315}{450}$$

$$\hat{p} = 0.70$$

Continue...

Step 5: Calculate Z-value: Formula: $Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Substitute values:

$$Z = \frac{0.75 - 0.66}{\sqrt{0.70(0.30)\left(\frac{1}{200} + \frac{1}{250}\right)}}$$

$$Z = \frac{0.09}{\sqrt{0.21(0.009)}}$$

$$Z = \frac{0.09}{0.0435}$$

$$Z = 2.07$$

Continue....

65

Step 6: Compare with Critical Value

Calculated value:

$$Z = 2.07$$

Critical values:

$$\pm 1.96$$

Since:

$$2.07 > 1.96$$

the value falls in the rejection region.

Step 7: Decision

Reject H_0 .

Step 8: Conclusion

At the 5% significance level, there is sufficient statistical evidence to conclude that the satisfaction proportions of the two companies are significantly different.

66

Thanks